

PRASAD ADHAU

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SUMMARY

Data Engineer with 4+ years of experience designing, building, and optimizing production-grade data pipelines and scalable cloud-native analytics platforms on GCP and AWS. Expert in Python, SQL, PySpark, and ETL/ELT pipelines, with a focus on data modeling, automation, CI/CD, and data quality, and building trusted data platforms that support analytics, KPI reporting, and ML workloads.

TECHNICAL SKILLS

Programming & ML: Python, C++, SQL, JavaScript, PySpark, Shell-scripting, Linux

Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn, scikit-learn, Tableau, Power BI

Databases & Big Data: MySQL, NoSQL, Data Warehousing, ETL/ELT Pipelines, Apache Airflow, Databricks, dbt

Cloud & DevOps: GCP, AWS, GitHub, CI/CD Pipelines, Process Automation, and Monitoring

Certifications: GCP Associate Cloud Engineer, SQL for Data Science, Coursera

EXPERIENCE

Teaching Assistant

August 2024 – December 2025

Rochester Institute of Technology

Rochester, NY

- Mentored 30+ students in databases, **data modeling**, and **EDA**, improving assignment completion by 25%.
- Streamlined grading processes for assignments and projects, reducing professor workload by 15 hours weekly.
- Conducted structured office hours weekly to guide students with course material and programming tools effectively.

Senior Engineer

March 2024 – June 2024

Nagarro

Pune, India

- Resolved critical data discrepancies across large-scale healthcare data pipelines, improving downstream reporting accuracy by 25%.
- Collaborated and mentored a team of 3 engineers to design and optimize **ETL workflows** for faster, reliable performance.
- Coordinated **cross-departmental** initiatives, reducing data issue resolution time from 8 to 3 hours weekly.

Data Engineer

December 2020 – March 2024

Accenture

Mumbai, India

- Architected **Python/PySpark ELT** pipelines on BigQuery to support daily sales reporting, ensuring 99% data accuracy.
- Designed dimensional data models and optimized schemas across **big data** ecosystems to enable scalable analytics.
- Constructed **Tableau** dashboards, reducing manual reporting by 40% and delivering faster, insight-driven visualizations.
- Optimized **SQL** queries and Python **automation scripts**, improving data retrieval time by 30% and saving 10+ hours weekly.
- Developed end-to-end **data warehousing** pipeline using **Talend, AWS, and Apache Airflow**, reducing ingestion time by 45%.
- Established **CI/CD** processes and data quality frameworks throughout the development lifecycle for seamless source data ingestion.
- Deployed **data validation, unit tests, and automated monitoring** alerts to ensure pipeline reliability and SLA management.

PROJECTS

Amazon Review Sentiment Analyzer | ML, NLP, GUI

January 2025 – May 2025

- Engineered **NLP pipeline** with BERT, DistilBERT, and Hugging Face, achieving 92% sentiment accuracy on 50K+ reviews.
- Delivered aspect-based sentiment analysis using Random Forest, TF-IDF, and embeddings, achieving 0.85 R² score.
- Produced interactive GUI with Matplotlib visualizations, processing 1000+ reviews with sub-second inference using Python.

Flight Network Dashboard | Neo4j, JavaScript (D3.js), Node.js

February 2025 – April 2025

- Created an interactive **Neo4j** dashboard using **JavaScript (D3.js)** and **Node.js** to visualize 3.5K+ airports and 46K+ routes.
- Implemented dynamic Cypher query execution and JavaScript-based data filtering, enabling instant, user-driven graph updates.
- Enhanced visualization clarity with custom JavaScript features like zoom, pan, hover tooltips, and export, improving usability by 40%.

Prediction of Early Readmission of Diabetes Patients | ML Classification, Ensemble Methods

October 2024 – December 2024

- Built ensemble classification model (**Random Forest, XGBoost**) on 15K+ patient records, achieving 88% accuracy.
- Executed **ML pipeline** with SMOTE, feature selection, hyperparameter tuning, and 5-fold cross-validation for optimization.
- Performed comprehensive evaluation and model comparison across 4 algorithms for optimal clinical prediction performance.

EDUCATION

Rochester Institute of Technology

Rochester, NY

Master of Science in Information Technology and Analytics, GPA: 3.85/4.0

August 2024 - Graduating May 2026

- Award:** Debbie Cahn Memorial Scholarship Awardee

Yeshwantrao Chavan College of Engineering

Nagpur, India

Bachelor of Engineering in Information Technology

August 2016 – June 2020

PUBLICATION & AWARDS

- Reasoning LLM-based Security Policy Generation for the I2NSF Framework - Published at KICS Winter Conference 2026 (International collaboration with SKKU, South Korea)
- Certificate of Excellence in BigQuery, Accenture | Self-Starter Award for Client Value Creation, Accenture